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Post-Hurricanes Roadway Closure Detection using Satellite Imagery and Semi-Supervised Ensemble Learning

--Manuscript Draft--

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Abstract

After hurricanes, roadway damage assessment is critical to emergency responders and city authorities. In this paper, we propose an automated semi-supervised approach to identify tree debris along roadways to improve the efficiency of damage assessment and allow faster debris cleaning operations. The solution uses two high-resolution satellite images taken before and after a hurricane. The proposed methodology is an ensemble learning machine using an unsupervised autoencoder-based feature extractor, an unsupervised vegetation coverage estimator, and a weakly-supervised tree segmentation. We show that such a combination can increase precision and accuracy. Our solution has been tested with a case study on Hurricane Michael, which hit Tallahassee, the capital of Florida, in October 2018.

Keywords: Remote Sensing, Tree Debris Detection, Post-Hurricane Assessment

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1 Introduction

Remote sensing for detecting changes on Earth's surface has become an essential topic in the recent decades with a wide range of applications, spanning both urban planning, agriculture investigation, environmental monitoring, and disaster assessment. Especially the disaster assessment application is an essential task that should be performed after catastrophic events like hurricanes, earthquakes, and tsunamis.

Hurricanes, in particular, have a significant impact in the US. The National Oceanic and Atmospheric Administration (NOAA) estimates a total cost of $1.75 trillion for weather and climate disasters damages[1] from 1980 to 2019 with the United States being among the five countries most hit by natural disasters [2]. In the recent years, Florida was hit by three destructive hurricanes: Hurricane Hermine (2016), Irma (2017), and Michael (2018) causing major infrastructure damages such as fallen trees, roadway closures due to tree debris and power outages in the entire peninsula [3]. Disaster evaluation, in most cases, is performed manually by visual inspection, which requires the emergency response crews and volunteers to drive around the affected area to map and assess damages. Such operation is time-consuming and very risky, especially in the proximity of fallen trees and power lines, and collapsed buildings. The increasing availability of satellites in orbit, providing imagery from anywhere on Earth with a high revisiting time and decreasing cost, bring new hopes to monitor such areas from space. Image processing techniques and the most recent artificial intelligence (AI) algorithms can reduce the need for visual inspection and make the entire assessment process fast and automated.

In the literature, some studies have already investigated post-disaster damage assessment using satellite imagery caused by tsunamis [4], [5], earthquakes [6] and floods [7]. Regarding tree debris assessment after extreme weather events, most of the works deal with estimating tree failure probability models in relation to strong winds. For example, [8] and [9] propose an empirical model to estimate the possibility of trees failures. [10] proposes a tree fragility model for an individual tree subject to wind storms. Such works often try to predict the tree crown and other estimated parameters [11] to combine them with wind data. Although such models are useful for better assessing the consequences of fallen trees, very few works have been done to detect fallen trees. [12] and [13] propose an integrated method for detecting fallen trees using airborne LiDAR point clouds. LiDAR is a valuable tool in remote sensing applications as it can generate a reliable 3D representation of the environment with high accuracy. However, its usage makes it also very expensive for mapping large areas (for example, an entire city), and the generated data are heavy and intensive to process. [14] uses traditional image processing filters, based on edge detection and color filtering, to detect tree debris using aerial imagery. The developed tool is rapid and relatively reliable but, according to the authors, the algorithm has difficulties in distinguishing downed trees from some other urban features such as fences around urban structures and shadows caused by buildings. Furthermore, aerial images cannot be acquired as easily as satellite images. [15] estimated stand-level tree volumes by averaging their reflectance values using Landsat satellite and regression analysis. [16] also used Landsat data as part of their assessment in estimating hurricane debris following the 2004–2005 Florida hurricane season. However, these works deal with large scale images, useful to have an overview of the most hit areas and evaluate forest damages but are not able to show individual downed trees on the roadways at the city level. At this level, it is often challenging to evaluate if a group of trees is simply covering the road or if they are lying on the pavement after been stroked by a hurricane. To better distinguish between the different situations, a change detection analysis is needed.

Change detection is the process of identifying differences in the state of an object or the land use by observing it at different times [17]. Recently, several techniques have been proposed to detect changes in satellite images. In [18], the authors propose an automatic approach to detect damaged buildings using image classification built on convolutional neural networks (CNNs). [19] proposes a supervised change detection architecture based on U-Nets [20]. In [21] the authors propose a supervised architecture based on CNNs to separate trivial changes from non-trivial ones. The drawback of most of these approaches is the need for complete datasets to train such networks, which are often unavailable or very limited and expensive. This lack of labeled data is problematic because most machine-learning algorithms are supervised learning methods that need a lot of labeled data for proper performance. Since obtaining a large number of labeled samples for supervised training is usually time-consuming and labor-intensive, many efforts have been made to achieve AI-based change detection in an unsupervised or semi-supervised manner [22], [23]. Usually, features are extracted with unsupervised AI models and the final change map is then generated by using a clustering algorithm [24], [25], [26] [27]. In [28] the authors use an autoencoder (AE) model to learn the transformation for a couple of co-registered images. In [29] a deep architecture based on CNN and autoencoders is proposed for change detection in image

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time series. Although fully unsupervised change detection algorithms do not require labeled training samples, sometimes the lack of semantic information or domain expertise as prior knowledge about the change we want to detect (for example, fallen trees) makes them unsuitable for the specific tasks. The available unsupervised machine learning techniques do not easily solve this problem [22].

In this paper, we focus on roadway closures after hurricanes due to fallen trees. We use high-resolution satellite images taken before and after a hurricane to perform change detection analysis and automatically locate areas where trees have landed on the roadways. Our proposed framework is semi-supervised. It is based on a combination of a fully unsupervised anomaly detection algorithm based on autoencoders, an unsupervised vegetation coverage detection, and a weakly-supervised tree segmentation algorithm. This approach alleviates the burden of obtaining extensive hand-labeled data sets (required for fully-supervised approaches). It can solve the issues related to a lack of prior knowledge typical of fully-unsupervised methods. To the authors' knowledge, no previous work has been done using the framework we present in this paper. The solution can help emergency responders and other transportation authorities immediately identify critical areas and create a map of the debris along the roadways. The outcomes of the proposed satellite-based framework will improve the damage assessment and hurricane restoration.

The novelties and contributions of our paper are as follows:

- Detect tree debris on roadways' level using high-resolution satellite images.
- Develop a semi-supervised ensemble method for tree debris detection. Such method merges the benefits of fully-unsupervised methods (no need of extensive labeled data) with prior semantic information coming from weakly-supervised tree segmentation. Results show the improved performances of our ensemble strategy.

2 Study Area and Data Description

Hurricane Michael was one of the strongest storms hitting the Southeast coast of the United States. It made landfall as an unprecedented Category 5 hurricane in the Florida Panhandle region with maximum sustained wind speeds of 140 knots (161 mph) bringing catastrophic storm surge to the Florida State and Big Bend areas (especially Mexico Beach and Panama City) [30]. It hit the City of Tallahassee on October 10th 2018, leaving 1.2 million electricity customers without power in several east coasts and southern states [31]. Estimated damage from Michael throughout the United States reached $25 billion [32].

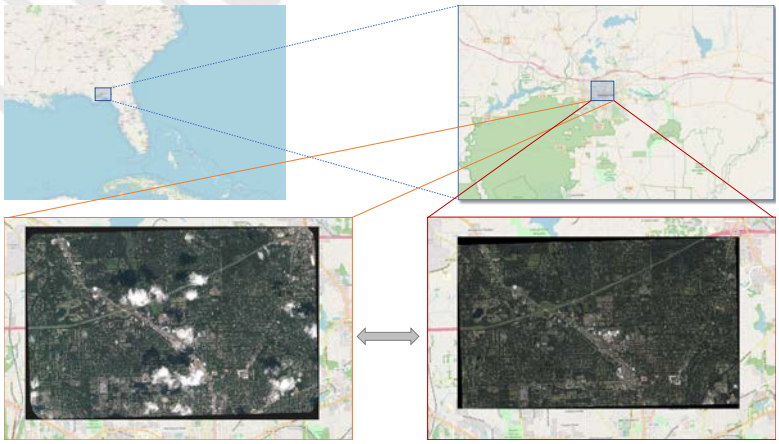

Figure 1: Study area in Tallahassee (Florida) where satellite images before and after Hurricane Michael are acquired

As a medium-sized city and the state capital, Tallahassee has a population of 193,551 as of the year 2018. It is also the home for two major universities, namely Florida State University (FSU) and Florida Agricultural and Mechanical University (FAMU). As a result, students comprise more than 35% of the entire population [33]. More than 30 state government agency headquarters, including the Capitol Building, Florida Supreme Court and Florida Governor's Mansion are located in Tallahassee. As such, the City of Tallahassee was selected for the proposed case study.

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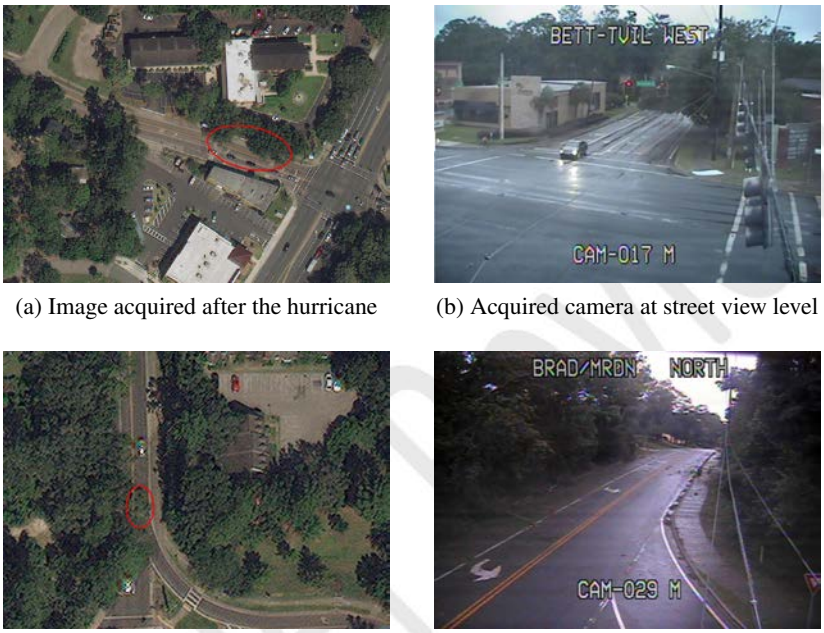
We acquired two high-resolution satellite images on a large portion of Tallahassee (**Figure** 1) before and after the hurricane. Images are provided by Maxar WorldView satellites, as described in **Table** 1.

Table 1: Satellite images used in this paper

	Date	Channels	Resolution
Pre-event image	September 14 2018	RGB-NIR	0.5m/pixel
Post-event image	October 13 2018	RGB-NIR	0.3m/pixel

Figure 2 shows some examples between an acquired satellite image and a fallen tree at the roadway level.



(a) Image acquired after the hurricane (b) Acquired camera at street view level

(c) Image acquired after the hurricane (d) Acquired camera at street view level

Figure 2: Comparison between the satellite image with an image acquired from a traffic camera ([35])

For validation purposes, we use the datasets related to roadway closures and tree debris locations achieved from the City of Tallahassee through the DigiTally mobile app [34]. This app enables the residents to directly connect with the City of Tallahassee and report power outages or request a service. **Figure** 3 shows the locations of tree debris that have been stacked before being collected by the city authorities in the considered area.

Figure 3: Map showing the locations of tree debris acquired from DigiTally App

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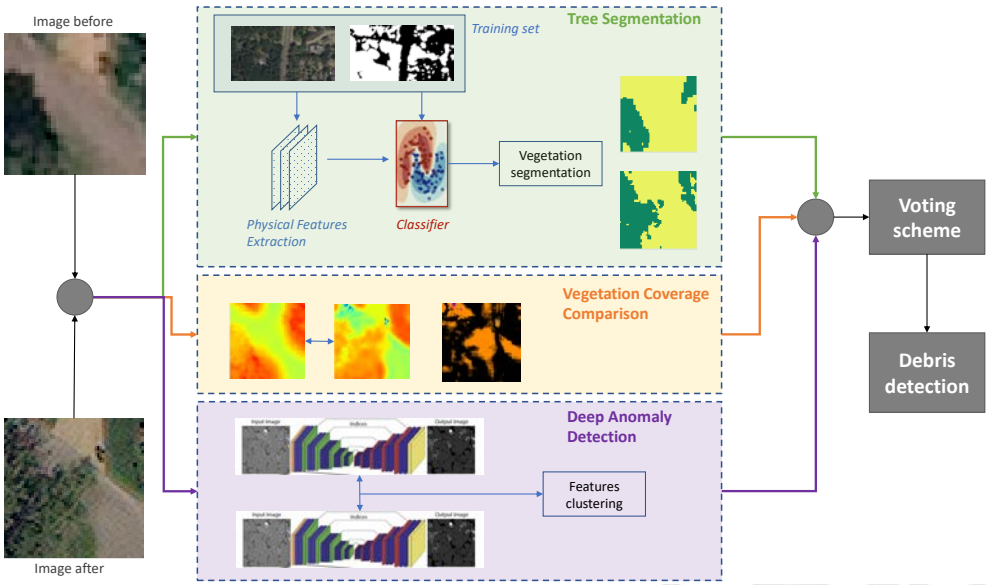

Figure 4: Overview of the proposed vegetation detection framework

3 Methodology

In this paper, our goal is to detect fallen trees along roadways using an automated way. Thus, we propose a semi-supervised approach with no need to have a labeled dataset of fallen trees to train the model. This will make it practical and suitable for real-world applications. The framework relies on a combination of an unsupervised anomaly detection model with vegetation-based estimation techniques, as shown in **Figure** 4.

To create the semi-supervised ensemble learning approach for detecting roadway closure, we implement three blocks, each independent from each other: a tree segmentation, a vegetation coverage comparison, and deep anomaly detection. We then integrate the results with a voting scheme.

3.1 Tree Segmentation Block

The first block is the tree segmentation block. Given an image I as input, the output M is a binary black and white mask where each pixel is marked as white if it is part of a tree and black otherwise. A training dataset consisting of images and the corresponding tree segmentation map is needed to training the vegetation detection algorithm. This is the only part of our approach, where a labeled dataset is needed, which makes the overall framework semi-supervised. Such training dataset can have a small size.

Specific features are extracted from the satellite image and stacked into a vector associated with each pixel. We assign to such a vector to the corresponding label from a labeled image in the training set. Then, we train the machine learning classifier using these feature vectors to create a pixel-level binary segmentation. As features, we extract the color pixel intensity coming from the different channels of the satellite image. We compute texture signatures using the Gray Level Co-Occurrence Matrix (GLCM) matrix, [37]. Given an image, I, with L-bit depth ($p = 2^L$ values) shaped as $n \times m$ pixels the GLCM matrix P, with size $p \times p$ is defined:

$$P_{\Delta x,\Delta y}(i,j) = \sum_{x=1}^{n}\sum_{x=1}^{m} \begin{cases} 1, & I(x,y) = i \wedge I(x+\Delta x, y+\Delta y) = j \\ 0, & \text{otherwise} \end{cases} \tag{1}$$

where (i,j) are the pixel values, (x,y) are the spatial positions in the image I, $(\Delta x, \Delta y)$ refers to the chosen offset for which this matrix is calculated (equal to (1,0) and (0,1) in this paper), and $I(x,y)$ indicates the pixel value at pixel (x,y).

From such a matrix, specific texture operators [38] can be extracted. Although a large number of operators exist, most of them are correlated as explained in [39]. Therefore, we decide to use only contrast and correlation operators defined as:

– *Contrast:*

$$\sum_{i,j=0}^{p} P_{i,j}(i-j)^2 \tag{2}$$

– *Correlation:*

$$\sum_{i,j=0}^{p} P_{i,j}\frac{(i-\mu_i)(j-\mu_j)}{\sqrt{\sigma_i^2\sigma_j^2}} \tag{3}$$

where

$$\mu_i = \sum_{i,j=0}^{N-1} i(P_{i,j}) \qquad \mu_j = \sum_{i,j=0}^{N-1} j(P_{i,j})$$
$$\sigma_i^2 = \sum_{i,j=0}^{N-1} P_{i,j}(i-\mu_i)^2 \qquad \sigma_j^2 = \sum_{i,j=0}^{N-1} P_{i,j}(j-\mu_j)^2$$

The next feature we use in this study is a Gaussian kernel. The Gaussian kernel (with variance equal to 1) is convolved with a gray-scale image derived from the RGB components. Such a kernel acts as a low-pass filter leading to a slightly blurred image. The Gaussian low pass filters are being more prevalent in image processing to cancel the noise [40].

The computed features values are stacked into a vector and we use the AdaBoost ensemble technique [41], [42] as a classifier. AdaBoost consists of a combination of individual weak learners and mathematically, can be expressed as:

$$H(x) = \sum_{i=1}^{T} \alpha_i h_i(x) \tag{4}$$

where $H(x)$ corresponds to the prediction, i is an iteration index, and T is the number of weak classifiers. $h_i(x)$ is a decision stump (one-leaf decision tree) with internal threshold and weighting factor α_i. Both the threshold and α_i are learned during the AdaBoost training step. The magnitude of $H(x)$ gives a confidence score of the prediction while the sign of $H(x)$ gives the prediction itself (i.e., tree or no-tree). A sigmoid function is used to map the interval $H(x) \in (-\infty, +\infty)$ into $(0,1)$:

$$P_{tree}(x) = \frac{1}{1+e^{-H(x)}} \tag{5}$$

Once the classifier is trained, predicting the binary mask from a new image is straightforward. One needs to extract the same feature vectors used during the training stage from the image and apply **Equation** (4). $H(x)$ is then used to assign a label to such vectors. Since a vector is computed for every pixel, a black and white mask can readily be created.

As pixel-wise classification is intrinsically noisy, we develop a refinement process based on energy minimization and for which energy can be defined as:

$$E = \sum_{i\in I} D(x_i) + \sum_{i,j\in\mathcal{N}} K|x_i - x_j| \tag{6}$$

where E is the energy, $P_{tree}(x_i)$ is computed from **Equation** (5) and $D(\cdot)$ is defines as follows:

$$D(x_i) = \begin{cases} P_{tree}(x_i) & \text{if } x_i = 0 \\ 1 - P_{tree}(x_i) & \text{if } x_i = 1 \end{cases} \tag{7}$$

x_i is the value of the pixel in position i in the image I, and j are all pixels belonging to the neighborhood $\mathcal{N}$ of i. The value of function $D(x_i)$ is low when one assigns the label 0 to black pixels or the label 1 to white pixels, and the value of $D(x_i)$ is high otherwise. The second term in **Equation** (6) measures the local smoothness of labeling with K, a regularization term (the higher the value, the smoother the refinement). We, then, solve the minimization problem by the max-flow/min-cut graph-based algorithm, which is a popular choice in the literature [44] for image processing applications. **Figure** 5 shows an visual example of the segmentation workflow.

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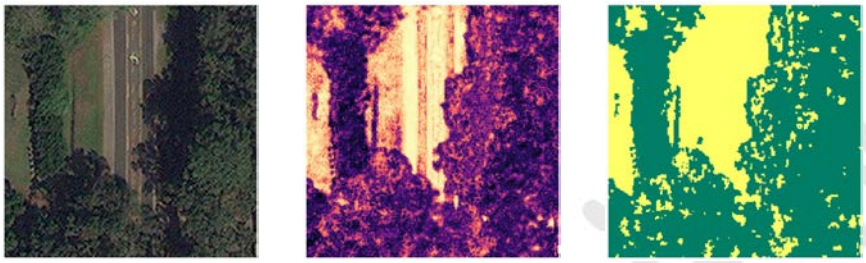
Finally, to quantify a similarity between two predicted segmentation maps, a dissimilarity score $\mathscr{S}_{tree}$ is computed as the integral over the window W of the pixel-wise difference between the two maps multiplied by a Gaussian kernel K centered at the center of the window (**Equation** 8).

$$\mathscr{S}_{tree} = \iint_W |M_a - M_b| \otimes K \tag{8}$$

In this way, large differences of pixels close to the center are weighted more than the ones near the edges since, for the task involved, we are more interested in changes occurring at the center of the window. From **Equation** (8) we note that the more M_b and M_a are similar, the lower the value of $\mathscr{S}_{tree}$ is.



(a) RGB image (b) Probabilistic output of the classifier (c) Final binary map

Figure 5: Example of a tree segmentation

3.2 Vegetation Coverage Comparison Block

The second block uses the Normalized Difference Vegetation Index (NDVI) [36] to recognize vegetation. NDVI is a popular tool in remote sensing for vegetation detection and defined as:

$$NDVI = \frac{\rho_{nir} - \rho_{red}}{\rho_{nir} + \rho_{red}} \tag{9}$$

where ρ_{red} and ρ_{nir} stand for the spectral reflectance measurements acquired in the red (visible) and near-infrared regions, respectively. Green living plants look brighter in the near-infrared band due to chlorophyll near-infrared high reflectance. In a similar way to the previous block, we define a dissimilarity score $\mathscr{S}_{NDVI}$ as the integral over the window of the difference between after and before, where only positive values are considered, with a multiplication with a Gaussian kernel K:

$$\mathscr{S}_{NDVI} = \iint_W \left(NDVI_a - NDVI_b \right) \Big|_+ \otimes K \tag{10}$$

Figure 6 shows a visual example of the comparison between NDVIs.

3.3 Deep Anomaly Detection Block

The third block is a fully unsupervised approach that compares the two images before and after to detect if significant changes occurred. Since comparing the raw images pixel-wisely is not feasible, we use convolutional autoencoders (AEs) [45] as a deep feature extractor with a Siamese configuration [46]. The use of convolutional autoencoders has been motivated by the strong capabilities of deep learning methods to learn useful features in a low-dimensionality space. It is similar to performing Principal Value Decomposition (PCA) in images without dealing with the lack of training data, likewise other supervised methods. An autoencoder is trained to learn a representational of each image in a reduced space (dimensionality reduction) and features are extracted from the bottleneck layer (**Figure** 7).

An autoencoder is mainly composed by two parts: an encoder and a decoder. The encoder encodes the input $\mathbf{x}$ to get latent features $h(\mathbf{x})$ and the decoder reconstructs the learned latent features to output a vector $\tilde{\mathbf{x}}$ that should be as close as possible to the original $\mathbf{x}$. Mathematically it can be formulated as:

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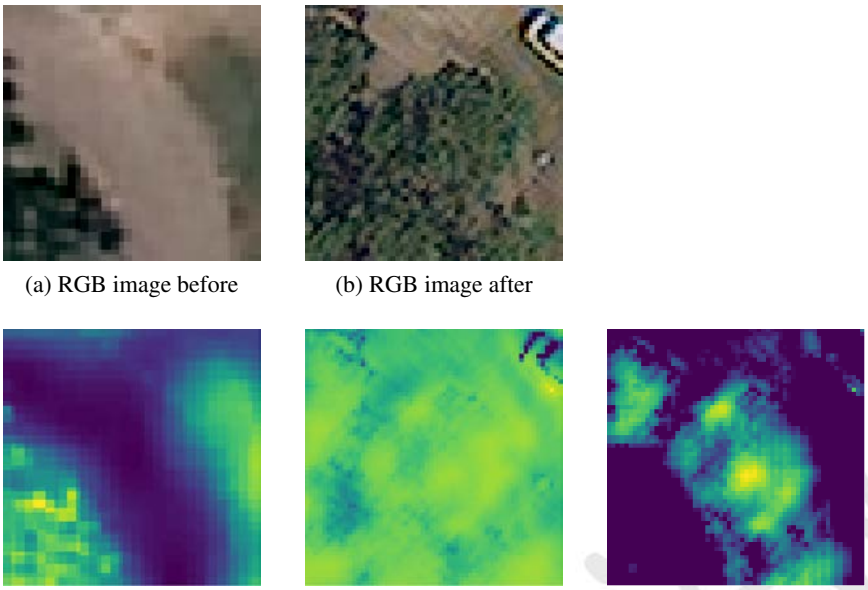

(a) RGB image before (b) RGB image after

(d) NDVI before (e) NDVI after (f) Difference

Figure 6: Example of a NDVI comparison

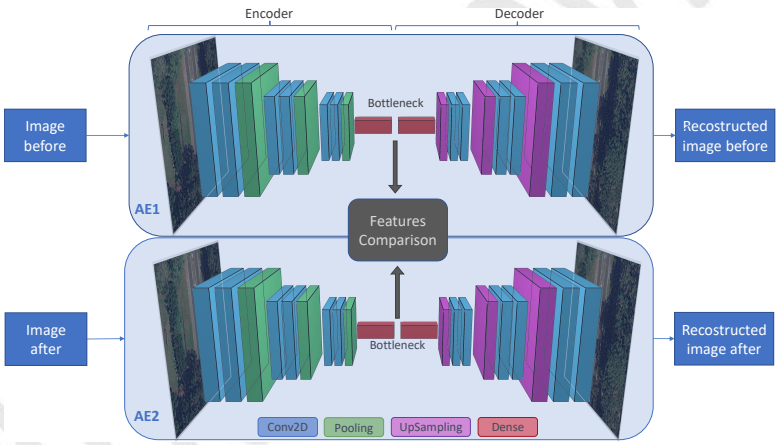

Figure 7: Two autoencoders are combined in a Siamese architecture and used as feature extractors, such features are then compared

$$\text{encoder:} \quad \mathbf{x} \rightarrow f(\mathbf{W}\mathbf{x}+\mathbf{b}) = \mathbf{z} \tag{11}$$

$$\text{decoder:} \quad \mathbf{z} \rightarrow h(\mathbf{W}'\mathbf{z}+\mathbf{b}') = \tilde{\mathbf{x}} \tag{12}$$

where $\mathbf{W}, \mathbf{W}', \mathbf{b}$ and $\mathbf{b}'$ are trainable parameters and can be obtained through unsupervised schemes. One autoencoder (AE_1) is trained to learn features from the *before* images while the other one (AE_2) is trained with the *after* images (see **Figure** 7).

We use the same idea as the previous block to calculate a dissimilarity score of $\mathscr{S}_{AE}$. AE stands for autoencoder between the two images (before and after the hurricane). We compute the difference between features extracted by the bottleneck of AE_1 and the features extracted by AE_2. Such difference is then evaluated using an *Isolation Forest* algorithm [47].

Isolation forest is a tree-structure based unsupervised learning algorithm for anomaly detection, enabling isolating anomalous points in a dataset. The main advantage of this approach is the possibility of exploiting sampling techniques to the extent that it is not allowed to the other profile-based methods, creating a fast algorithm with a low memory demand [48].

3.4 Voting Scheme

Since each block runs independently from each other they can be combined together as follows:

- *Mode 1*: a location is marked as an anomaly if at least one of the three blocks detect an anomaly
- *Mode 2:* a location is marked as an anomaly if at least two of the three blocks detect an anomaly
- *Mode 3:* a location is marked as an anomaly if all the three blocks detect an anomaly

Figure 8 shows the different strategies.

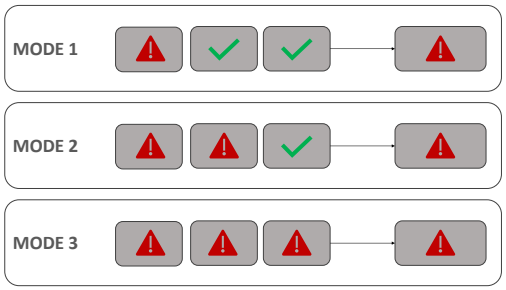

Figure 8: Different combination schemes: the location is marked as anomaly if there is agreement between a certain number of blocks, depending on the particular chosen voting scheme

4 Results

For validation purposes, two satellite images have been acquired for a portion of the City of Tallahassee (Florida), as described in Section 2. We use an 18×18 meters sliding window (corresponding to 60×60 pixels given the satellite resolution) along the roadway's path to extract patches from the two different satellite images, I_b, I_a, namely before and after the hurricane respectively. Patches are spatially spaced no more than 10 meters from each other. In this way, a total of 5,139 single windows are extracted covering the whole considered area. D_b is the dataset containing all the patches acquired before the hurricane, while D_a is the dataset containing all the patches acquired after the hurricane. Due to the different resolution, I_b has been up-sampled to match with the number of pixels of I_a.

The three blocks have been implemented as described in subsections 3.1, 3.2 and 3.3. The two autoencoders, AE_1 and AE_2, have been trained with the architecture described in **Table** 2 with mean squared error (MSE) as loss function between the input and reconstructed images, the first with D_b and the second with D_a. With the proposed architecture, the raw input RGB image ($3 \times 60 \times 60 = 10800$ pixels) is encoded into only 450 values ($2 \times 15 \times 15$). The dataset is split into training and validation sets with a 20% ratio. The model has been trained for 20 epochs, in **Figure** 9 the loss plot for the training and validation sets is showed. We note that the loss function stabilizes after 10 epochs, and there is no over-fitting. **Figure** 10 shows some examples of reconstructed images.

Table 2: Autoencoder architecture

Encoder	Decoder
Input(60,60,3)	UpSampling(2)
Conv2D (3,3,8) + Relu	Conv2D (3,3,8) + Relu
MaxPooling(2)	UpSampling(2)
Conv2D (3,3,2) + Relu	Conv2D (3,3,3) + Sigmoid
MaxPooling(2)	*Output(60,60,3)*

For each patch, dissimilarity scores $\mathscr{S}_{tree}$, $\mathscr{S}_{NDVI}$ and $\mathscr{S}_{AE}$ are calculated and normalized into the interval $(0,1)$. **Figure** 11 shows the histogram of these values for all the 5139 patches.

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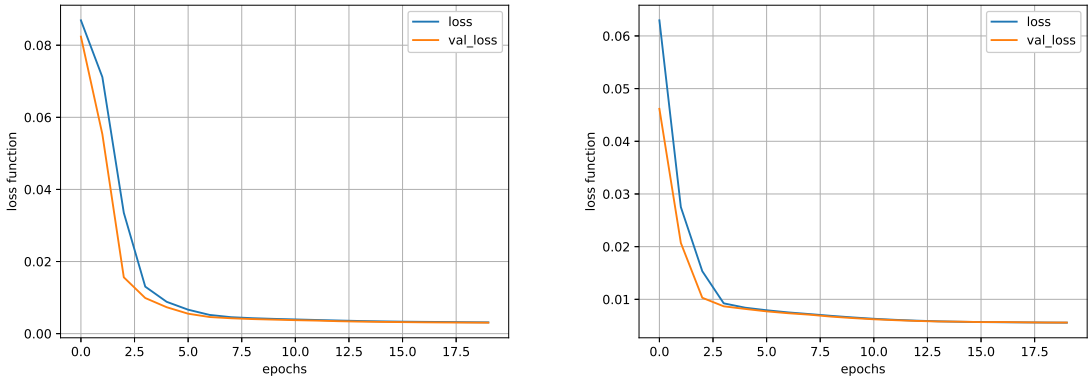


(a) Training history for D_b

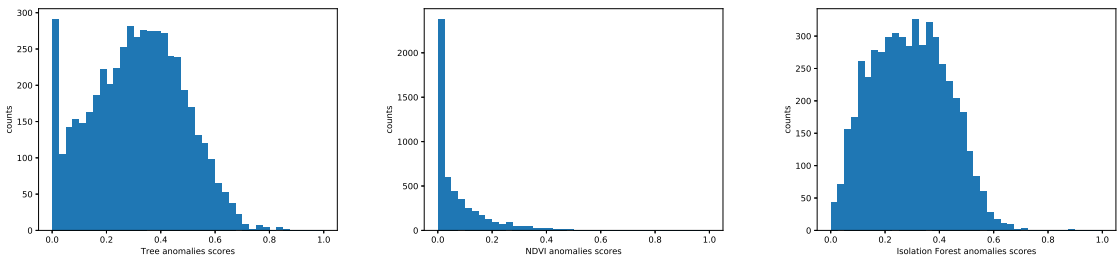
(b) Training history for D_a

Figure 9: Training loss function plot for D_b (dataset with images before the hurricane) and D_a ((dataset with images after the hurricane))

(a) D_b: original images (*first row*), reconstructed images by the trained autoencoder (*second row*)

(b) D_a: original images (*first row*), reconstructed images by the trained autoencoder (*second row*)

Figure 10: Example of reconstructed images.



(a) Histogram of $\mathscr{S}_{tree}$

(b) Histogram of $\mathscr{S}_{NDVI}$

(c) Histogram of $\mathscr{S}_{AE}$

Figure 11: Histogram of the dissimilarities values for each block. From left to right: $\mathscr{S}_{tree}$, $\mathscr{S}_{NDVI}$ and $\mathscr{S}_{AE}$

It is noted that the distributions' profile makes sense because most of the locations present intermediate changes. The more we move to the right (higher numbers mean more substantial differences between the I_b and I_a), the fewer counts in the histogram we have. Theoretically, if we had an increasing distribution, it would mean that there are more anomalies (changes) than unchanged locations, and the whole anomaly detection idea would be ill-posed. Nevertheless, we are interested in locations whose dissimilarity values are statistically larger than the mean. Therefore, a threshold is implemented. Given the distribution $\mathscr{D}$ we get from the histogram, in this paper a threshold value Th is computed as:

$$Th = \mu_{\mathscr{D}} + \sigma_{\mathscr{D}} \tag{13}$$

where $\mu_{\mathscr{D}}$ is the mean and $\sigma_{\mathscr{D}}$ the standard deviation of the distribution $\mathscr{D}$. In this way, if the pair (I_b, I_a) has a dissimilarity value larger than the threshold, we say that there is an anomaly between the two images.

To validate the results, we use the tree debris locations of the area obtained from the DigiTally mobile application, as described in Section 2. Such locations show the stacked piles of tree debris before their collection by the city operators. However, their precision on the map is not perfect and not always match with fallen trees that are visually observable on the roadway. To solve this issue and have a meaningful and fair comparison, we introduce a distance threshold of 20 meters. The accuracy scores, True Positive (TP), True Negative (TN), False Positive (FP) and False Negative (FN), are calculated as sketched in **Figure** 12.

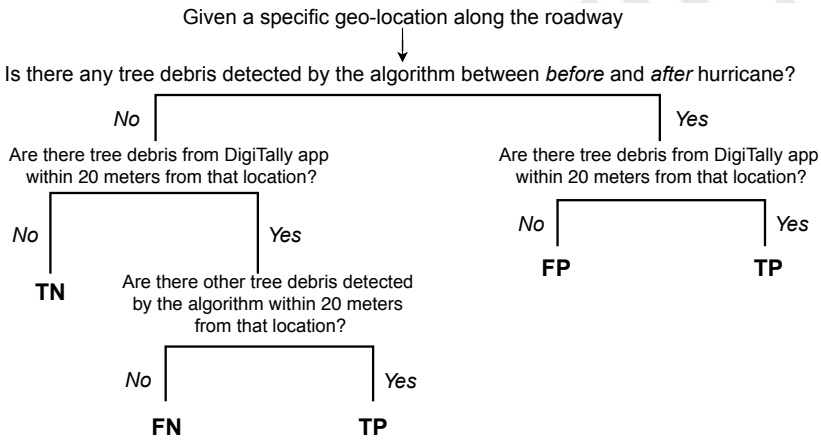


Figure 12: Flowchart to calculate True/False Positive and True/False Negative

Accuracy and precision are calculated as:

$$Precision = \frac{TP}{TP+FP} \quad Accuracy = \frac{TP+TN}{P+N} \tag{14}$$

Finally, we use **Equations** (14) to evaluate the performances of each block separately as well as the different combinations proposed in Section 3.4, results are shown in **Table** 3.

From **Table** 3, we see that each block separately has similar scores. Since they are fully independent of each other, they can significantly boost precision when they are combined. The *Mode 1* has a lower precision because we are merging all the anomalies detected in each block, and it seems most of them are false positives leading to less precision. On the other hand, the *Mode 3* has the highest precision but also lower accuracy than the different modes. The observation is valid because we are selecting only the anomalies detected by all three blocks at the same time, so true positives are high. However, one block can have difficulties in detecting anomalies that can easily be observed by the other two. This voting scheme will not include this anomaly, even if it should be supposed to be considered as such, and it will increase the false negatives, leading to a drop in the accuracy. *Mode 2* reaches the best tradeoff. We call a location has an anomaly or, in other words, has a difference in trees condition before and after the hurricane if at least two blocks of ensemble learning machine have detected it. Our algorithm automatically detects up to 78% of the debris without the need to have a labeled dataset for fallen trees.

Finally, the framework's output is used to create a heat map showing all the detected debris within the area, as shown in **Figure** 13. This map is created automatically scanning all the roadways, the benefit and usefulness of having, in a fully automated and unsupervised way, such a map for the city authorities is straightforward.

Table 3: Accuracy scores

Mode	Accuracy	Precision
Tree segmentation	0.758	0.655
NDVI comparison	0.726	0.637
Deep anomaly detection	0.762	0.657
Mode 1	0.724	0.551
Mode 2	**0.781**	0.812
Mode 3	0.685	**0.897**

Figure 13: Example of heat map showing all the detected anomalies within the area

4.1 Discussions and lessons learned

The results show that satellite images are valuable tools for post-disaster assessment. However, some limitations are intrinsic in the analytical process. Comparing different satellite images from different times, providers, and capturing angles are is challenging in nature. Although an ortho-rectification procedure is performed as a pre-processing step, minor changes in tilt and orientation of images can lead to shifts in objects (see **Figure** 14).

Satellite imageries can also be acquired by different sensors, meaning they will have different resolutions and spectral reflectances. Furthermore, shadows are a source of mistake since they change dynamically due to the sun and earth position. Tree's canopies are also subject to change due to wind speed and gust. Furthermore, large trees at the roadside cover the pavement surface from the top view. The issues mentioned above make the tree debris detection on roadways from satellite images a challenging task. Follows are some examples of problematic pieces of satellite images for detecting fallen trees on roads. **Figure** 15 shows the shadows issue for tree detection.

Sometimes, it is not easy to match the location of fallen trees and the debris report by the DigiTally application, see **Figure** 16. The stacked debris might be removed before or after the image is recorded. The other issue is due to cloud coverage that blocks the available satellite images from commercial providers. Therefore, it is not always possible to acquire an image at the next day after the hurricane.

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(a) image before (b) image after

Figure 14: Small shift between two acquired images: note how the shapefile location showing the road slightly changes between the two images

(a) image before (b) image after

Figure 15: Shadows make the assessment evaluation more difficult

(a) image before (b) image after

Figure 16: Some areas with debris points look very similar and is difficult to locate if and where fallen trees are actually present, even if according to the DigiTally App there are (red dots in the middle)

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5 Conclusions

In this paper, we developed an automated semi-supervised framework to detect fallen trees at roadway's level precision. We used two satellite images of a portion of Tallahassee (Florida), acquired before and after Hurricane Michael. Fallen trees and debris in the middle of the road pavement are considered anomalies. We present three completely different and independent methods. The first is based on weakly-supervised tree segmentation to locate trees, the second one detects differences in vegetation coverage using spectral information while the third is a fully unsupervised, state-of-the-art deep learning algorithm. We show that combining these methods can boost the overall accuracy and precision in detecting fallen trees. Despite the challenging task, our solution shows promising results and is also practical to implement since it does not require a labeled dataset with fallen trees. The proposed solution can generate a map of the areas showing the detected critical points. Such a map can be quickly generated right after the event and improve and ease the restoration procedure conducted by city and state agencies.

As future work, we aim to investigate the current anomaly detection techniques to make the approach more robust. Also, automatic roadways detection can be performed, making the framework more applicable even when roadway shapefiles are not available.

6 Author Contributions

The authors confirm contribution to the paper as follows: study conception and design: M. Gazzea, A. Karaer, N. Balafkan, E. E. Ozguven, and R. Arghandeh; analysis and interpretation of results: M.Gazzea, A. Karaer, N. Balafkan, and R. Arghandeh; manuscript preparation: M. Gazzea, A. Karaer, N. Balafkan, E. E. Ozguven and R. Arghandeh. All authors reviewed the results and approved the final version of the manuscript.

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